

⇒ Picard Method

The Picard method, also called the Picard iteration method, is a way to find the approximate solutions to differential equations. It is mainly used to prove that solutions to certain types of differential eq exist and to improve accuracy step by step.

so →

Formula:-

$$y_n = y_0 + \int_{n_0}^n f(n, y_{n-1}) dn$$

ex: solve

$$\frac{dy}{dn} = 1 + ny \quad \text{with} \quad n_0 = 0, y_0 = 0$$

$$y_n = y_0 + \int_{n_0}^n f(n, y_{n-1}) dn$$

$$y_1 = y_0 + \int_{n_0}^n (1 + y_{0-1}) dn$$

$$y_1 = y_0 + \int_{n_0}^n (1 + n \cdot 0) dn$$

$$y_1 = 0 + \int_{n_0}^n dn$$

$$y_1 = [n]_{n_0}^n$$

$$y_1 = n - 0 \quad \boxed{y_1 = n}$$

$$y_2 = y_0 \int_{n_0}^n f(n, y_{2-1}) dn.$$

$$y_2 = 0 \int_{n_0}^n (1 + n \cdot n) dn.$$

$$y_2 = \int_{n_0}^n (1 + n^2) dn.$$

$$y_2 = \left[n + \frac{n^3}{3} \right]_{n_0}^n$$

$$\boxed{y_2 = n + \frac{n^3}{3}}$$

now third iteration:-

$$y_3 = 0 \int_{n_0}^n f(n, y_{3-1}) dn.$$

$$y_3 = \int_{n_0}^n \left(1 + n + \frac{n^3}{3} \right) dn.$$

$$y_3 = \left[n + \frac{n^2}{2} + \frac{n^4}{4} \right]_{n_0}^n$$

$$y_3 = \int_{n_0}^n \left(1 + n \left[n + \frac{n^3}{3} \right] \right) dn.$$

$$y_3 = \int_{n_0}^n \left(1 + n^2 + \frac{n^4}{3} \right) dn$$

$$y_3 = \left[n + \frac{n^3}{3} + \frac{n^5}{3 \cdot 5} \right]_{n=0}^n$$

$$y_3 = \left[n + \frac{n^3}{3} + \frac{n^5}{15} \right]_{n=0}^n$$

$$y_3 = \left[n + \frac{n^3}{3} + \frac{n^5}{15} \right] \text{ Ans.}$$

⇒ ex 2 class work

$$\frac{dy}{dn} = y + n$$

$$\boxed{y_0 = 1, n_0 = 0}$$

→ formula

$$y_n = y_0 + \int_{n_0}^n f(n, y_{n-1}) dn.$$

sol

$$y_1 = 1 + \int_{n_0}^n (n + 1) dn.$$

$$y_1 = 1 + \left[\frac{n^2}{2} + n \right]_{n_0}^n.$$

$$\boxed{y_1 = 1 + \frac{n^2}{2} + n} \rightarrow \text{first iteration:}$$

→ Now for second iteration:-

$$y_2 = y_0 + \int_{n_0}^n f(n, y_{n-1}) dn.$$

$$y_2 = 1 + \int_{n_0}^n n \left[1 + \frac{n^2}{2} + n \right] dn.$$

$$y_2 = 1 + \int_{n_0}^n 2n + \frac{n^3}{2} + n^2 dn$$

$$y_2 = 1 + \frac{2n^2}{2} + \frac{n^3}{6} + n^2$$

$$\boxed{y_2 = 1 + n^2 + \frac{n^3}{6} + n^2} \text{ Ans.}$$

$\Rightarrow$ Now solve $\frac{dy}{dn} = y + n$ using linear differential eq

sol
 $\frac{dy}{dn} - y = n.$

$$\frac{dy}{dn} (-1)(y) = (n.)$$

$$P(n) = -1 \quad Q(n) = n$$

$$I.P.M. = e^{\int -1 dn} = e^{-n} \rightarrow \text{Integral factor}$$

$$y(e^{-n}) = \int n e^{-n} \rightarrow \text{Solve by using Integral factor of U.V.}$$

$$\left\{ \begin{array}{l} U \cdot V = U \int r dr - \int \frac{du}{dn} \cdot \int r dr. \\ \text{P.L.A.T.E.} \end{array} \right.$$

$$\left[\text{Let } U = x \quad r = e^{-n} \right]$$

$$= n \frac{e^{-n}}{-1} - \int 1 \cdot \frac{e^{-n}}{-1}$$

$$= n \frac{e^{-n}}{-1} + \frac{e^{-n}}{-1} + C.$$

$$y e^{-n} = -n e^{-n} - e^{-n} + C$$

$$y = -n \frac{e^{-n}}{e^{-n}} - \frac{e^{-n}}{e^{-n}} + \frac{C}{e^{-n}}$$

$$\boxed{y = -n - 1 + C e^n} \quad \text{General solution:}$$

Given that $y(0) = 1 \quad n_0 = 0.$

$$1 = -0 - 1 + C e^0 \rightarrow \boxed{y = -n - 1 + 2e^n}$$

$1 = \frac{C - 1}{e - 1}$